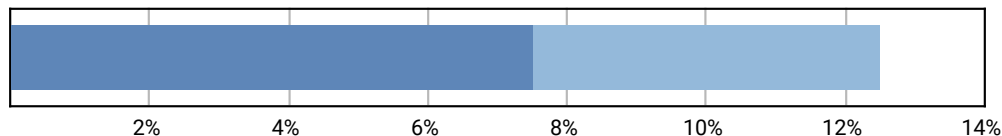


ADV: Statistics (Adv), S3 Random Variables (Adv)
Probability Density Functions (Y12)

Teacher: Ljiljana Todic

Exam Equivalent Time: 108 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

S3 Normal Distributions and Probability Density Functions



***Analytics based on the average contribution to Advanced HSC exams since 2020.**

■ Normal Distributions
 ■ Probability Density Functions

HISTORICAL CONTRIBUTION

- *S3 Random Variables* is a Year 12 topic that hasn't previously existed in the Mathematics/Advanced course. It has attracted eye-opening allocations of 14 and 11 marks in the first two new syllabus Advanced exams, over 4 separate questions in each exam!
- The topic is split into two sub-topics for the purposes of analysis: *1-Normal Distribution (7.5%)* and *2-Probability Density Functions (5.0%)*.
- This analysis looks at *Probability Density Functions*.

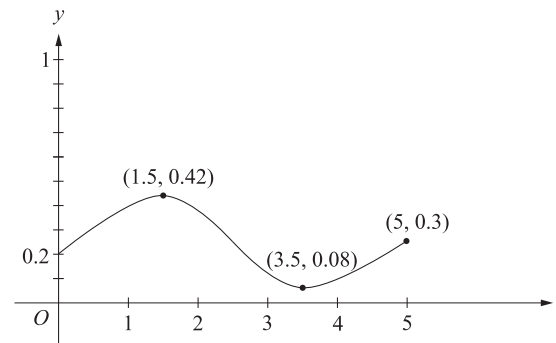
HSC ANALYSIS - What to expect and common pitfalls

- *Probability Density Functions* is a sub-topic that has not appeared in any HSC mathematics course previously.
- The 2021 and 2020 Advanced exams allocated 6 and 4 marks to this topic respectively, producing sub-50% mean marks on each occasion it was examined.
- Students must be able to find the median and mode given a probability density function. The latter was examined in *2021 Q24* and caused major problems, producing a mean mark of 24%. Finding these and other probability percentiles using cumulative distributive functions is a core skill well covered in the database.
- Students must have a deep understanding of the link between a cumulative distributive functions and their probability density functions, most recently tested in *2021 Q30* with a mean mark of just 22%. This question type can attract a significant mark allocation and must be a revision focus (see *EQ-Bank 1 and 16*).

Questions

1. Statistics, 2ADV S3 EQ-Bank 6 MC

A continuous probability density function graph is drawn below.



Which of the following is the mode?

- 0.42
- 0.08
- 1.5
- 5

2. Statistics, 2ADV S3 SM-Bank 5 MC

The function $f(x)$ is a probability density function of a continuous random variable with the rule

$$f(x) = \begin{cases} ae^x & 0 \leq x \leq 1 \\ ae & 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The value of a is

- 1
- $\frac{1}{e}$
- $\frac{1}{2e}$
- $\frac{1}{2e - 1}$

3. Statistics, 2ADV S3 SM-Bank 6 MC

The continuous random variable, $\mathbf{X}$, has a probability density function given by

$$f(x) = \begin{cases} \frac{1}{4}\cos\left(\frac{x}{2}\right) & 3\pi \leq x \leq 5\pi \\ 0 & \text{elsewhere} \end{cases}$$

The value of $\mathbf{a}$ such that $P(X < a) = \frac{\sqrt{3} + 2}{4}$ is

- A. $\frac{19\pi}{6}$
- B. $\frac{14\pi}{3}$
- C. $\frac{10\pi}{3}$
- D. $\frac{29\pi}{6}$

4. Statistics, 2ADV S3 SM-Bank 7 MC

A probability density function $\mathbf{f(x)}$ is given by

$$f(x) = \begin{cases} \frac{1}{12}(8x - x^3) & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

The median $\mathbf{m}$ of this function satisfies the equation

- A. $-m^4 + 16m^2 - 6 = 0$
- B. $m^4 - 16m^2 = 0$
- C. $m^4 - 16m^2 + 24 = 0.5$
- D. $m^4 - 16m^2 + 24 = 0$

5. Statistics, 2ADV S2 SM-Bank 14

A probability density function $\mathbf{f(x)}$ is given by

$$f(x) = \begin{cases} px(3-x) & \text{if } 0 \leq x \leq 3 \\ 0 & \text{if } x < 0 \text{ or if } x > 3 \end{cases}$$

where $\mathbf{p}$ is a positive constant.

- i. Find the value of $\mathbf{p}$. (2 marks)
- ii. Find the mode of $\mathbf{f(x)}$. (2 marks)

6. Statistics, 2ADV S2 SM-Bank 15

A function $\mathbf{f(x)}$ is given by

$$f(x) = \begin{cases} \frac{3}{4}(x-2)(4-x) & \text{if } 2 \leq x \leq 4 \\ 0 & \text{if } x < 2 \text{ or if } x > 4 \end{cases}$$

- i. Show this curve is a probability density function. (2 marks)
- ii. Find the mode. (2 marks)

7. Statistics, 2ADV S3 SM-Bank 20

A continuous random variable $\mathbf{X}$ has a probability density function given by

$$f(x) = \begin{cases} Cx + D & 2 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

where $\mathbf{C}$ and $\mathbf{D}$ are constants.

Find the exact values of $\mathbf{C}$ and $\mathbf{D}$, given the median of $\mathbf{X}$ is 4. (4 marks)

8. Statistics, 2ADV S3 SM-Bank 10

A continuous random variable, $\mathbf{X}$, has a probability density function given by

$$f(x) = \begin{cases} \frac{1}{5}e^{-\frac{x}{5}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The median of $\mathbf{X}$ is $\mathbf{m}$.

Determine the value of $\mathbf{m}$. (3 marks)

9. Statistics, 2ADV S3 SM-Bank 11

The probability density function of a continuous random variable $\mathbf{X}$ is given by

$$f(x) = \begin{cases} \frac{x}{12} & 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbf{P(X < 3)}$ (2 marks)

10. Statistics, 2ADV S3 SM-Bank 12

The function

$$f(x) = \begin{cases} k \sin(\pi x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is a probability density function for the continuous random variable $\mathbf{X}$.

Show that $\mathbf{k = \frac{\pi}{2}}$. (2 marks)

11. Statistics, 2ADV S3 SM-Bank 2

If a continuous random variable $\mathbf{X}$ has probability density function

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the exact value of $\mathbf{p}$ such that $\mathbf{P(X > p) = 0.4}$. (3 marks)

12. Statistics, 2ADV S3 SM-Bank 8

The continuous random variable $\mathbf{X}$ has a distribution with probability density function given by

$$f(x) = \begin{cases} ax(5-x) & \text{if } 0 \leq x \leq 5 \\ 0 & \text{if } x < 0 \text{ or if } x > 5 \end{cases}$$

where $\mathbf{a}$ is a positive constant.

a. Find the value of $\mathbf{a}$. (3 marks)

b. Express $\mathbf{P(X < 3)}$ as a definite integral. (Do **not** evaluate the definite integral.) (1 mark)

13. Statistics, 2ADV S3 SM-Bank 9

The probability density function $\mathbf{f(x)}$ of a random variable $\mathbf{X}$ is given by

$$f(x) = \begin{cases} \frac{x+1}{12} & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of $\mathbf{b}$ such that $\mathbf{P(X \leq b) = \frac{5}{8}}$. (3 marks)

14. Statistics, 2ADV S3 2020 HSC 23

A continuous random variable, $\mathbf{X}$, has the following probability density function.

$$f(x) = \begin{cases} \sin x & \text{for } 0 \leq x \leq k \\ 0 & \text{for all other values of } x \end{cases}$$

a. Find the value of $\mathbf{k}$. (2 marks)

b. Find $\mathbf{P(X \leq 1)}$. Give your answer correct to four decimal places. (2 marks)

15. Statistics, 2ADV S3 EQ-Bank 16

A probability density function is defined by

$$f(x) = \begin{cases} a & \text{for } 0 \leq x \leq 4 \\ 3a & \text{for } 4 < x \leq 8 \end{cases}$$

i. Find the value of $\mathbf{a}$. (2 marks)

ii. Sketch the probability density function. (1 mark)

iii. Find an expression for the cumulative distribution function. (2 marks)

16. Statistics, 2ADV S3 EQ-Bank 1

A probability density function can be used to model the lifespan of a termite, $\mathbf{X}$, in weeks, is given by

$$f(x) = \begin{cases} k(36 - x^2) & 3 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- Show that the value of k is $\frac{1}{45}$. (2 marks)
- Find the cumulative distribution function. (2 marks)
- Find the probability that a termite's lifespan is greater than 5 weeks. (1 mark)

17. Statistics, 2ADV S3 SM-Bank 18

The Lorenz birdwing is the largest butterfly in a habitat.

The probability density function that describes its life span, $\mathbf{X}$, in weeks, is given by

$$f(x) = \begin{cases} \frac{4}{625}(5x^3 - x^4) & 0 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

In a sample of 80 Lorenz birdwing butterflies, how many butterflies are expected to live longer than two weeks, correct to the nearest integer? (2 marks)

18. Statistics, 2ADV S3 2021 HSC 30

The number of hours for which light bulbs will work before failing can be modelled by the random variable $\mathbf{X}$ with cumulative distribution function

$$F(x) = \begin{cases} 1 - e^{-0.01x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Jane sells light bulbs and promises that they will work for longer than exactly 99% of all light bulbs.

Find how long, according to Jane's promise, a light bulb bought from her should work. Give your answer in hours, rounded to two decimal places. (2 marks)

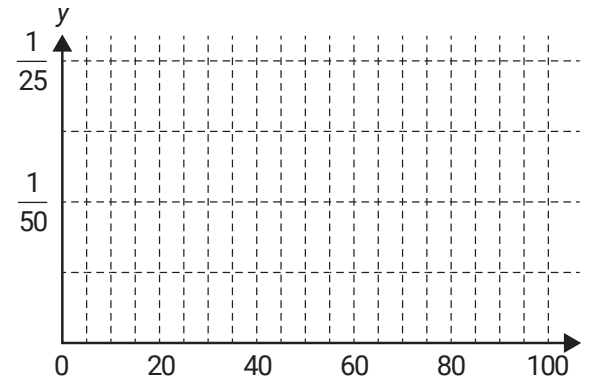
19. Statistics, 2ADV S3 SM-Bank 13

The time Jennifer spends on her homework each day varies, but she does some homework every day.

The continuous random variable $\mathbf{T}$, which models the time, $\mathbf{t}$, in minutes, that Jennifer spends each day on her homework, has a probability density function $\mathbf{f}$, where

$$f(t) = \begin{cases} \frac{1}{625}(t - 20) & 20 \leq t < 45 \\ \frac{1}{625}(70 - t) & 45 \leq t \leq 70 \\ 0 & \text{elsewhere} \end{cases}$$

- Sketch the graph of $\mathbf{f(t)}$ on the axes provided below. (3 marks)



- Find the mode. (1 mark)
- Find $P(25 \leq T \leq 55)$. (2 marks)

20. Statistics, 2ADV S3 SM-Bank 3

The continuous random variable $\mathbf{X}$ has a probability density function given by

$$f(x) = \begin{cases} \pi \sin(2\pi x) & \text{if } 0 \leq x \leq \frac{1}{2} \\ 0 & \text{elsewhere} \end{cases}$$

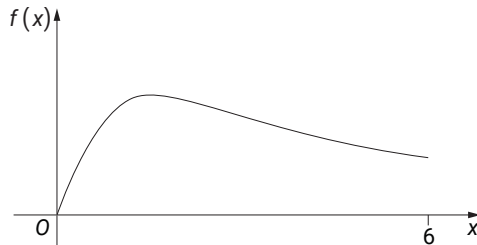
Find the value of $\mathbf{a}$ such that $P(\mathbf{X} > \mathbf{a}) = 0.2$. Give your answer to 2 decimal places. (3 marks)

21. Statistics, 2ADV S3 2021 HSC 33

People are given a maximum of six hours to complete a puzzle. The time spent on the puzzle, in hours, can be modelled using the continuous random variable $\mathbf{X}$ which has probability density function

$$f(x) = \begin{cases} \frac{Ax}{x^2 + 4} & \text{for } 0 \leq x \leq 6, \text{ (where } A > 0) \\ 0 & \text{for all other values of } x \end{cases}$$

The graph of the probability density function is shown below. The graph has a local maximum.



a. Show that $A = \frac{2}{\ln 10}$. (2 marks)

b. Show that the mode of $\mathbf{X}$ is two hours. (2 marks)

c. Show that $P(\mathbf{X} < 2) = \log_{10} 2$. (2 marks)

d. The Intelligence Quotient (IQ) scores of people are normally distributed with a mean of 100 and standard deviation of 15.

It has been observed that the puzzle is generally completed more quickly by people with a high IQ.

It is known that 80% of people with an IQ greater than 130 can complete the puzzle in less than two hours.

A person chosen at random can complete the puzzle in less than two hours.

What is the probability that this person has an IQ greater than 130? Give your answer correct to three decimal places. (2 marks)

22. Statistics, 2ADV S3 SM-Bank 4

The continuous random variable $\mathbf{X}$ has a probability density function given by

$$f(x) = \begin{cases} \cos(2x) & \text{if } \frac{3\pi}{4} < x < \frac{5\pi}{4} \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of $\mathbf{a}$ such that $P(\mathbf{X} < \mathbf{a}) = 0.25$. Give your answer correct to 2 decimal places. (3 marks)

Worked Solutions

1. Statistics, 2ADV S3 EQ-Bank 6 MC

Mode is the most common x -value.

y -axis measures the probability.

Highest probability = 0.42

$$\therefore \text{Mode} = 1.5$$

$$\Rightarrow C$$

2. Statistics, 2ADV S3 SM-Bank 5 MC

Total area = 1

$$\int_0^1 ae^x dx + \int_1^2 ae dx = 1$$

$$[ae^x]_0^1 + [ae \cdot x]_1^2 = 1$$

$$[ae - a] + [2ae - ae] = 1$$

$$2ae - a = 1$$

$$a(2e - 1) = 1$$

$$\therefore a = \frac{1}{2e - 1}$$

$$\Rightarrow D$$

Worked Solutions

3. Statistics, 2ADV S3 SM-Bank 6 MC

$$\int_{3\pi}^a \frac{1}{4} \cos\left(\frac{x}{2}\right) dx = \frac{\sqrt{3} + 2}{4}$$

$$\left[\frac{1}{2} \sin\left(\frac{x}{2}\right) \right]_{3\pi}^a = \frac{\sqrt{3} + 2}{4}$$

$$\frac{1}{2} \left[\sin\left(\frac{a}{2}\right) - \sin\left(\frac{3\pi}{2}\right) \right] = \frac{\sqrt{3} + 2}{4}$$

$$\frac{1}{2} \sin\left(\frac{a}{2}\right) + \frac{1}{2} = \frac{\sqrt{3} + 2}{4}$$

$$\sin\left(\frac{a}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\frac{a}{2} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \dots$$

$$\therefore a = \frac{14\pi}{3} \quad (3\pi \leq a \leq 5\pi)$$

$$\Rightarrow B$$

4. Statistics, 2ADV S3 SM-Bank 7 MC

$$\int_0^m \frac{1}{12} (8x - x^3) dx = 0.5$$

$$\left[\frac{1}{12} \left(4x^2 - \frac{x^4}{4} \right) \right]_0^m = 0.5$$

$$4m^2 - \frac{m^4}{4} = 6$$

$$16m^2 - m^4 = 24$$

$$m^4 - 16m^2 + 24 = 0$$

$$\Rightarrow D$$

5. Statistics, 2ADV S2 SM-Bank 14

i. Total Area under curve = 1

$$p \int_0^3 3x - x^2 dx = 1$$

$$p \left[\frac{3}{2}x^2 - \frac{x^3}{3} \right]_0^3 = 1$$

$$p \left[\frac{27}{2} - \frac{27}{3} - 0 \right] = 1$$

$$\frac{9p}{2} = 1$$

$$p = \frac{2}{9}$$

ii. $f(x) = \frac{2}{9}(3x - x^2)$

$$f'(x) = \frac{2}{9}(3 - 2x)$$

S.P. when $f'(x) = 0$:

$$3 - 2x = 0$$

$$x = \frac{3}{2}$$

$$f''(x) = -\frac{4}{9} < 0 \Rightarrow \text{MAX}$$

$$\therefore \text{Mode} = \frac{3}{2}$$

6. Statistics, 2ADV S2 SM-Bank 15

i. $f(x) = \frac{3}{4}(x - 2)(4 - x)$

$$\int_2^4 f(x) = \frac{3}{4} \int_2^4 -x^2 + 6x - 8 dx$$

$$= \frac{3}{4} \left[\frac{-x^3}{3} + 3x^2 - 8x \right]_2^4$$

$$= \frac{3}{4} \left[\left(-\frac{64}{3} + 48 - 32 \right) - \left(-\frac{8}{3} + 12 - 16 \right) \right]$$

$$= \frac{3}{4} \left(-\frac{16}{3} + \frac{20}{3} \right)$$

$$= 1$$

$$f(2) = 0, f(4) = 0$$

$$f(x) > 0 \text{ for } 2 < x < 4$$

$$f(2) \geq 0 \text{ for } 2 \leq x \leq 4$$

$\therefore f(x)$ is a probability density function.

ii. $f(x) = \frac{3}{4}(-x^2 + 6x - 8)$

$$f'(x) = \frac{3}{4}(-2x + 6)$$

$$f''(x) = -\frac{3}{2}$$

SP when $f'(x) = 0$

$$-2x + 6 = 0$$

$$x = 3$$

$$f''(x) < 0 \Rightarrow \text{MAX}$$

$$\therefore \text{Mode} = 3$$

7. Statistics, 2ADV S3 SM-Bank 20

$$\int_2^5 Cx + D \, dx = 1$$

$$\left[\frac{C}{2}x^2 + Dx \right]_2^5 = 1$$

$$\left[\left(\frac{25}{2}C + 5D \right) - (2C + 2D) \right] = 1$$

$$\frac{21}{2}C + 3D = 1$$

$$21C + 6D = 2 \dots (1)$$

Using median $X = 4$:

$$\left[\frac{C}{2}x^2 + Dx \right]_2^4 = 0.5$$

$$[(8C + 4D) - (2C + 2D)] = 0.5$$

$$6C + 2D = 0.5 \dots (2)$$

Multiply: $(2) \times 3$

$$18C + 6D = 1.5 \dots (3)$$

Subtract: $(1) - (3)$

$$3C = \frac{1}{2}$$

$$\therefore C = \frac{1}{6}$$

Substitute into (1):

$$\frac{21}{6} + 6D = 2$$

$$6D = -\frac{9}{6}$$

$$\therefore D = -\frac{1}{4}$$

8. Statistics, 2ADV S3 SM-Bank 10

$$\frac{1}{5} \int_0^m e^{-\frac{x}{5}} \, dx = \frac{1}{2}$$

$$\frac{1}{5} \times (-5) \left[e^{-\frac{x}{5}} \right]_0^m = \frac{1}{2}$$

$$\left[-e^{-\frac{x}{5}} \right]_0^m = \frac{1}{2}$$

$$-e^{-\frac{m}{5}} + 1 = \frac{1}{2}$$

$$e^{-\frac{m}{5}} = \frac{1}{2}$$

$$-\frac{m}{5} = \log_e \left(\frac{1}{2} \right)$$

$$\therefore m = -5 \log_e \left(\frac{1}{2} \right) \quad (\text{or } 5 \log_e(2), \text{ or } \log_e 32)$$

9. Statistics, 2ADV S3 SM-Bank 11

$$P(X < 3)$$

$$= \int_1^3 \frac{1}{12} x \, dx$$

$$= \frac{1}{12} \left[\frac{1}{2} x^2 \right]_1^3$$

$$= \frac{1}{24} [3^2 - 1^2]$$

$$= \frac{1}{3}$$

10. Statistics, 2ADV S3 SM-Bank 12

Total Area under curve = 1 u²

$$\int_0^1 k \sin(\pi x) dx = 1$$

$$-\frac{k}{\pi} [\cos(\pi x)]_0^1 = 1$$

$$-\frac{k}{\pi} [\cos(\pi) - \cos(0)] = 1$$

$$-\frac{k}{\pi} [-1 - 1] = 1$$

$$2k = \pi$$

$$\therefore k = \frac{\pi}{2}$$

11. Statistics, 2ADV S3 SM-Bank 2

$$\int_p^2 \frac{x}{2} dx = 0.4$$

$$\left[\frac{x^2}{4} \right]_p^2 = \frac{2}{5}$$

$$1 - \frac{p^2}{4} = \frac{2}{5}$$

$$\frac{p^2}{4} = \frac{3}{5}$$

$$p^2 = \frac{12}{5}$$

$$\therefore p = \frac{2\sqrt{3}}{\sqrt{5}}$$

$$= \frac{2\sqrt{15}}{5} \quad (0 \leq p \leq 2)$$

12. Statistics, 2ADV S3 SM-Bank 8

a. Total Area under curve = 1

$$a \int_0^5 (5x - x^2) dx = 1$$

$$a \left[\frac{5}{2}x^2 - \frac{1}{3}x^3 \right]_0^5 = 1$$

$$a \left[\left(\frac{125}{2} - \frac{125}{3} \right) - (0) \right] = 1$$

$$\frac{125}{6}a = 1$$

$$\therefore a = \frac{6}{125}$$

b. $P(X < 3) = \int_0^3 ax(5 - x) dx$

13. Statistics, 2ADV S3 SM-Bank 9

$$\frac{1}{12} \int_0^b (x + 1) dx = \frac{5}{8}$$

$$\left[\frac{1}{2}x^2 + x \right]_0^b = \frac{15}{2}$$

$$\frac{1}{2}b^2 + b = \frac{15}{2}$$

$$b^2 + 2b - 15 = 0$$

$$(b + 5)(b - 3) = 0$$

$$\therefore b = 3 \quad (0 \leq b \leq 4)$$

14. Statistics, 2ADV S3 2020 HSC 23

a. $\int_0^k \sin x = 1$

$$[-\cos x]_0^k = 1$$

$$-\cos k + \cos 0 = 1$$

$$-\cos k = 0$$

$$\cos k = 0$$

$$k = \frac{\pi}{2}$$

b. $P(X \leq 1) = \int_0^1 \sin x \, dx$

$$= [-\cos x]_0^1$$

$$= -\cos 1 + \cos 0$$

$$= 1 - \cos 1$$

$$= 0.45969\dots$$

$$= 0.4597 \text{ (to 4 d.p.)}$$

♦ Mean mark part (b) 44%.

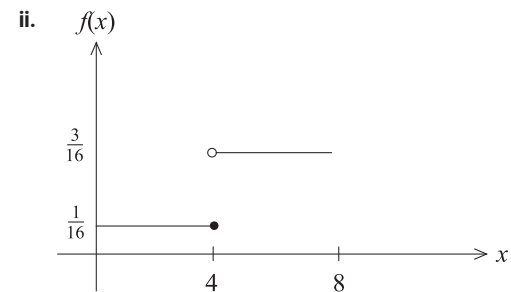
15. Statistics, 2ADV S3 EQ-Bank 16

i. $\int_0^4 a \, dx = [ax]_0^4 = 4a$

$$\int_4^8 3a \, dx = [3ax]_4^8 = 24a - 12a = 12a$$

$$4a + 12a = 1$$

$$a = \frac{1}{16}$$



iii. When $0 \leq x \leq 4$:

$$F(x) = \int \frac{1}{16} \, dx = \frac{x}{16} + C$$

$$F(0) = 0 \Rightarrow C = 0$$

$$F(x) = \frac{x}{16} \dots (1)$$

When $4 < x \leq 8$:

$$F(x) = \int \frac{3}{16} \, dx = \frac{3x}{16} + C$$

$$F(4) = \frac{1}{4} \text{ (see (1) above)}$$

$$\frac{3 \times 4}{16} + C = \frac{1}{4}$$

$$C = -\frac{1}{2}$$

$$F(x) = \frac{3x}{16} - \frac{1}{2}$$

COMMENT: Understand why you can check your equation here by confirming $F(8) = 1$

$$\therefore F(x) = \begin{cases} \frac{x}{16} & \text{for } 0 \leq x \leq 4 \\ \frac{3x}{16} - \frac{1}{2} & \text{for } 4 < x \leq 8 \end{cases}$$

16. Statistics, 2ADV S3 EQ-Bank 1

$$\text{i.} \quad k \int_3^6 36 - x^2 dx = 1$$

$$k \left[36x - \frac{x^3}{3} \right]_3^6 = 1$$

$$k[(216 - 72) - (108 - 9)] = 1$$

$$45k = 1$$

$$k = \frac{1}{45}$$

$$\text{ii.} \quad F(t) = \int_{-\infty}^t f(x) dx$$

$$= \int_3^t f(x) dx$$

$$= \frac{1}{45} \int_3^t 36 - x^2 dx$$

$$= \frac{1}{45} \left[36x - \frac{x^3}{3} \right]_3^t$$

$$= \frac{1}{135} [108x - x^3]_3^t$$

$$= \frac{1}{35} [(108t - t^3) - (324 - 27)]$$

$$= \frac{1}{135} (108t - t^3 - 297)$$

$$\therefore F(x) = \begin{cases} 0 & x < 3 \\ \frac{1}{135} (108t - t^3 - 297) & 3 \leq x \leq 6 \\ 1 & x > 6 \end{cases}$$

$$\text{iii.} \quad P(X > 5) = 1 - F(5)$$

$$= 1 - \frac{1}{135} (108 \times 5 - 5^3 - 297)$$

$$= 1 - \frac{118}{135}$$

$$= \frac{17}{135}$$

17. Statistics, 2ADV S3 SM-Bank 18

$$\begin{aligned}\Pr(X > 2) &= \frac{4}{625} \int_2^5 5x^3 - x^4 dx \\ &= \frac{4}{625} \left[\frac{5}{4} x^4 - \frac{x^5}{5} \right]_2^5 \\ &= \frac{4}{625} \left[\left(\frac{5^5}{4} - \frac{5^5}{5} \right) - \left(\frac{5}{4} \times 2^4 - \frac{2^5}{5} \right) \right] \\ &= \frac{4}{625} \left[\frac{625}{4} - \frac{68}{5} \right] \\ &= 0.9129 \dots\end{aligned}$$

$$\begin{aligned}\therefore \text{Expected number} &= 80 \times 0.9129 \dots \\ &\approx 73.03 \\ &\approx 73\end{aligned}$$

18. Statistics, 2ADV S3 2021 HSC 30

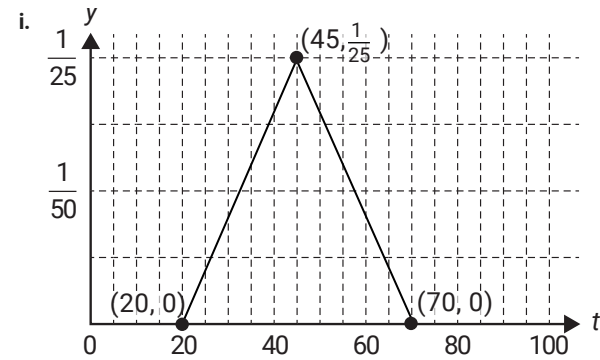
Find x when $F(x) > 0.99$:

$$\begin{aligned}1 - e^{-0.01x} &= 0.99 \\ e^{-0.01x} &= 0.01 \\ -0.01x &= \ln(0.01) \\ x &= \frac{\ln(0.01)}{-0.01} \\ &= 460.517 \dots \\ &= 460.52 \text{ hours}\end{aligned}$$

$\therefore$ Light bulbs should work at least 460.52 hours.

♦♦ Mean mark 22%.

19. Statistics, 2ADV S3 SM-Bank 13



ii. Mode = 45 (value of t at highest value of $f(t)$)

MARKER'S COMMENT: Many did not draw graph along t -axis between 0 and 20 and for $t > 70$

iii. $P(25 \leq T \leq 55)$

$$\begin{aligned}&= \int_{25}^{45} \frac{1}{625} (t - 20) dt + \int_{45}^{55} \frac{1}{625} (70 - t) dt \\ &= \frac{1}{625} \left[\frac{t^2}{2} - 20t \right]_{25}^{45} + \frac{1}{625} \left[70t - \frac{t^2}{2} \right]_{45}^{55} \\ &= \frac{1}{625} [112.5 - (-187.5)] + \frac{1}{625} (2337.5 - 2137.5) \\ &= \frac{300}{625} + \frac{200}{625} \\ &= \frac{4}{5}\end{aligned}$$

$$\int_a^{\frac{1}{2}} \pi \sin(2\pi x) dx = 0.2$$

$$-\frac{1}{2} [\cos(2\pi x)]_a^{\frac{1}{2}} = 0.2$$

$$-\frac{1}{2} [\cos(\pi) - \cos(2\pi a)] = 0.2$$

$$-\frac{1}{2} (-1 - \cos(2\pi a)) = 0.2$$

$$-1 - \cos(2\pi a) = -0.4$$

$$\cos(2\pi a) = -0.6$$

$$2\pi a = \cos^{-1}(-0.6)$$

$$\therefore a = \frac{\cos^{-1}(-0.6)}{2\pi}$$

$$= 0.3524...$$

$$= 0.35 \text{ (to 2 d.p.)}$$

$$\text{a. } \int_0^6 \frac{Ax}{x^2 + 4} dx = 1$$

$$\frac{A}{2} \int_0^6 \frac{2x}{x^2 + 4} dx = 1$$

$$\frac{A}{2} [\ln(x^2 + 4)]_0^6 = 1$$

$$\frac{A}{2} (\ln 40 - \ln 4) = 1$$

$$\frac{A}{2} \ln\left(\frac{40}{4}\right) = 1$$

$$\frac{A}{2} \ln 10 = 1$$

$$A = \frac{2}{\ln 10}$$

b. Mode $\rightarrow f(x)$ is a MAX

$$f(x) = \frac{Ax}{x^2 + 4}$$

$$f'(x) = \frac{A(x^2 + 4) - Ax(2x)}{(x^2 + 4)^2}$$

$$= \frac{Ax^2 + 4A - 2Ax^2}{(x^2 + 4)^2}$$

$$= \frac{A(4 - x^2)}{(x^2 + 4)^2}$$

$f(x)$ max occurs when $f'(x) = 0$:

$$4 - x^2 = 0$$

$$x = 2 \text{ (} x > 0 \text{)}$$

$$\text{c. } P(X < 2) = \int_0^2 \frac{Ax}{x^2 + 4} dx$$

$$= \frac{A}{2} [\ln(x^2 + 4)]_0^2$$

◆◆◆ Mean mark part (b) 24%.

◆◆ Mean mark part (c) 30%.

$$= \frac{1}{\ln 10} (\ln 8 - \ln 4)$$

$$= \frac{1}{\ln 10} \left(\ln \frac{8}{4} \right)$$

$$= \frac{1}{\ln 10} \cdot \ln 2$$

$$= \log_{10} 2$$

d. $z\text{-score}(130) = \frac{x - \mu}{\sigma} = \frac{130 - 100}{15} = 2$

$$P(z > 2) = 2.5\%$$

$$P(\text{IQ} > 130 \mid x < 2) = \frac{P(\text{IQ} > 130 \cap X < 2)}{P(X < 2)}$$

$$= \frac{0.8 \times 0.025}{\log_{10} 2}$$

$$= 0.0664 \dots$$

$$= 0.066$$

♦♦ Mean mark part (d) 25%.

22. Statistics, 2ADV S3 SM-Bank 4

$$\int_{\frac{3\pi}{4}}^a \cos(2x) \, dx = 0.25$$

$$\frac{1}{2} [\sin(2x)]_{\frac{3\pi}{4}}^a = 0.25$$

$$\left[\sin(2a) - \sin\left(\frac{3\pi}{2}\right) \right] = 0.5$$

$$\sin(2a) + 1 = 0.5$$

$$2a = \sin^{-1}(-0.5)$$

$$= -0.5235$$

Since sin is negative in 3rd/4th quadrants:

$$2a = \pi + 0.5234$$

$$a = 1.832 \dots \quad (\text{not in range})$$

or

$$2a = 2\pi - 0.5234$$

$$a = 2.87989 \dots$$

$$= 2.88 \quad \left(\text{in range: } \frac{3\pi}{4} < x < \frac{5\pi}{4} \right)$$